SE 216 – SOFTWARE PROJECT MANAGEMENT
REQUIREMENTS DOCUMENT

PROJECT NAME: StorySphere: A Safe and Interactive Digital Platform for Writers and Readers

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REQ. #	FUNCTIONAL REQUIREMENTS
1	The system must allow users to create an account by providing a valid email address and password, and must verify the email address using a confirmation link sent via email before account activation.
2	The system must allow registered users to log in and log out securely using their email and password, and must maintain user sessions using secure HTTP-only and SameSite cookies.
3	The system must verify users' age before granting access to +18 content by integrating with the e-Devlet Kapısı authentication system as follows: • Users must be redirected to the e-Devlet login page for identity verification • Users must authenticate using their Turkish national identity credentials • The system must retrieve the user's date of birth through the e-Devlet identity verification service • Access to +18 content must only be granted if the user is verified to be 18 years or older • The system must store only a verification flag and must not store national identity numbers or sensitive personal data
4	Authors must be able to create and edit personal profiles including biography, profile picture, and contact information.
5	Authors must be able to create, edit, and delete posts on their profiles.

6	Authors must be able to add, edit, and remove external book links associated with their profiles.
7	Authors must be able to host live streaming sessions using a real-time video streaming system based on WebRTC technology.
8	The system must provide real-time chat rooms using WebSocket-based communication to allow messaging between authors and readers.
9	Authors must be able to publish, edit, and delete stories using an integrated content editor.
10	The system must allow multiple authors to collaborate on a story by: • Supporting real-time collaborative editing using operational transformation or conflict-free replicated data types • Supporting asynchronous editing with version control and change history tracking
11	Readers must be able to highlight text within stories and save highlights as bookmarks linked to their accounts.
12	Readers must be able to rate stories using a numeric rating system from 1 to 5 and leave written reviews or comments.
13	The system must support a shared reading feature that allows multiple users to: • Join the same reading session via a shared session ID • View each other's reading progress in real time • Send reactions or messages during the session using a synchronized communication module

14	The system must support the following content formats: • Text-based stories displayed in a responsive reading interface • Manga-style content displayed as paginated images with zoom functionality • Audiobooks with playback controls including play, pause, skip, and progress tracking
15	The system must provide indexed search functionality using a search engine such as Elasticsearch, allowing filtering based on character name, profession, theme, and genre.
16	The system must provide automatic story translation between supported languages using neural machine translation models integrated via the OpenAI GPT and Google Translate API, and must allow users to select the target language before translation.
17	The system must provide writing suggestions during story creation using a custom-trained AI model integrated into the editor, and must support real-time sentence completion, grammar correction, and style improvement suggestions.
18	The system must provide an administrative moderation panel that allows administrators to review, approve, reject, or delete content flagged by the automated moderation system.
REQ. #	**NON-FUNCTIONAL REQUIREMENTS**

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1	The system must load all main pages within 2 seconds under normal network conditions with a minimum connection speed of 10 Mbps.
2	The system must maintain at least 99.9% uptime to ensure high availability.
3	The system must ensure secure authentication and data protection using: • HTTPS with TLS 1.3 encryption • AES-256 encryption for data at rest • bcrypt password hashing with salting and a minimum cost factor of 10 • Secure HTTP-only and SameSite cookies for session management
4	The system must handle at least 10,000 concurrent active users while ensuring that average response time does not increase by more than 20% under peak load conditions, using horizontally scalable cloud infrastructure.
5	The system must be compatible with: • Google Chrome version 120 and above • Mozilla Firefox version 120 and above • Microsoft Edge version 120 and above • Safari version 16 and above • Android version 10 and above • iOS version 15 and above
6	The system must: • Perform automatic full database backups daily at 02:00 AM (UTC+3) • Store backups in a secure cloud storage system • Retain backups for a minimum of 7 days • Support full system recovery within a maximum of 2 hours
7	The system must detect and filter inappropriate or offensive content by:

	• Using an external machine learning-based moderation service based on the OpenAI Moderation API • Performing real-time content analysis during content submission • Blocking or flagging content before publication if it violates predefined rules • Achieving at least 90% detection accuracy • Allowing flagged content to be reviewed and approved or rejected by administrators through a moderation dashboard
8	Search queries must return results within 3 seconds under normal system load conditions.
9	The system must prevent unauthorized data modification using role-based access control and server-side validation mechanisms.
10	The system must support multiple language options, including English, Turkish, Italian, French, Mandarin Chinese, Spanish, Arabic, Hindi, Portuguese, Russian, and German.
11	The system must allow users to adjust font size, background color, and enable dark mode.
12	The system must provide an intuitive user interface that allows new users to navigate and use core features within 5 minutes without external assistance.